Tushar Kanti Dey

Full-Stack Software Engineer — Scalable Systems Builder

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Professional Summary

Computer Science undergraduate with strong foundations in data structures, algorithms, and backend system design. Experienced in architecting full-stack applications with structured data models, secure authentication flows, and modular codebases. Focused on clean abstractions, performance optimization, and production-grade engineering practices.

Core Competencies

Frontend Engineering	Next.js, React.js, Tailwind CSS, TypeScript, Component Architecture	**Backend Engineering**	Node.js, NestJS, Express.js, RESTful API Design, Authentication Systems
Data Systems	PostgreSQL, Supabase, MongoDB, Redis, Schema Design	**Computer Science Foundations**	Data Structures, Algorithms, OOP, DBMS, Operating Systems
Mobile Systems	React Native, Expo, Rendering Optimization	**Engineering Tooling**	Git, GitHub, Docker, CI/CD, Vercel

Signature Projects

WebScope – Data Extraction and Processing System

Overview: Designed a backend-driven web data extraction platform focused on structured parsing, normalization, and API delivery.

Engineering Contributions:
- Implemented modular scraping pipelines with structured data transformation layers.
- Designed REST endpoints to securely serve processed data.
- Added validation, error isolation, and request handling safeguards.
- Structured codebase using separation of concerns for maintainability.

Tech Stack: Next.js, Node.js, REST APIs.

Fenix – Real-Time Communication Platform

Overview: Architected a secure real-time video communication system using WebRTC infrastructure.

Engineering Contributions:
- Integrated LiveKit for peer-to-peer video and audio streaming.
- Implemented authenticated session management using Clerk.
- Designed dynamic meeting lifecycle flows with unique room identifiers.
- Structured UI components for predictable rendering and scalability.

Tech Stack: Next.js, LiveKit, Clerk, Tailwind CSS.

Signifiya – Performance-Optimized Mobile Application

Overview: Built a production-ready mobile system emphasizing runtime efficiency and structured API integration.

Engineering Contributions:
- Reduced unnecessary component re-renders through state refactoring.
- Optimized build size via media compression and configuration tuning.
- Implemented secure environment-based API configuration handling.
- Managed Android build lifecycle and deployment workflows.

Tech Stack: React Native, Expo, REST APIs.

Taskhub – Modular Task Management System
Overview: Developed a modular task tracking system with structured CRUD workflows.
Engineering Contributions:
- Designed predictable state transitions for task lifecycle management.
- Built reusable component layers to improve maintainability.
- Structured UI layout for responsive, device-agnostic behavior.
- Deployed optimized production build via Vercel.

Tech Stack: Next.js, React, Tailwind CSS.

Education

2023–Present **B.Tech in Computer Science Engineering**, *Adamas University*, Kolkata, India

Key Strengths

- Strong grasp of algorithmic complexity and structured problem solving.
- Experience designing modular and maintainable system architectures.
- Clear understanding of relational schema design and API contracts.
- Comfortable debugging and optimizing production-level applications.